Public Catastrophic Crisis Briefing Assistant

Person 1: Data Sources, Validation, and Source-of-Truth Strategy

This document defines the data-source foundation for high-integrity crisis intelligence ingestion and verification.

Step 1: Data Sources | Step 2: Selection Table | Step 3: Validation | Step 4: Reasoning

Step 1 - Data Source Inventory

Primary objective: prioritize authoritative, machine-readable, frequently updated sources for catastrophic crisis signals.

Source	Domain	Update Pattern	Access Mode	Role
USGS Earthquake Hazards Program	Earthquakes, aftershocks	Near real-time event feed	Official API/feed	Primary for seismic events
NOAA / National Weather Service	Severe weather, alerts, warnings	Continuous operational updates	API + official products	Primary for weather catastrophes
FEMA Incident and Disaster resources	Declared disasters, response context	Operational + incident-cycle updates	Public data/services	Primary for U.S. federal incident context
CDC Emergency Preparedness & Response	Public health emergencies	Event-driven + periodic updates	Official bulletins/data pages	Primary for health-impact validation
WHO Emergencies	Global health and humanitarian emergencies	Situation updates and briefings	Official reports	Primary for international health context

Source	Domain	Update Pattern	Access Mode	Role
GDACS (Global Disaster Alert and Coordination System)	Multi-hazard global alerts	Rapid alerting + severity cues	Public feed/API	Primary global early-warning layer
ReliefWeb (UN OCHA)	Humanitarian reports and field updates	Frequent document-level updates	API + structured document metadata	Secondary corroboration and narrative context
NASA FIRMS	Wildfire hotspots and fire signals	Frequent satellite-derived updates	Data feed/API tools	Primary for wildfire signal intelligence
EM-DAT (historical disaster database)	Historical disaster reference	Periodic updates (not real-time operations)	Dataset access	Reference baseline; not operational real-time source

Step 2 - Selection Rationale Matrix

Decision table showing selected sources, alternatives considered, and backup strategy.

Source	Why Selected	Limitations	Backup / Reference Source	Why Backup is Useful
USGS	Authoritative seismic data with rapid updates and standardized event metadata.	Focused on seismic domain only.	GDACS	Cross-checks severity context and broader regional impact signals.
NOAA / NWS	Official weather warnings and watch/advisory	Weather-focused; not full multi-domain crisis coverage.	State emergency management bulletins	Adds local operational context and local

Source	Why Selected	Limitations	Backup / Reference Source	Why Backup is Useful
	structure trusted for public safety operations.			emergency actions.
FEMA	Federal disaster declarations and response status improve policy and action clarity.	Declaration timelines may lag onset signals.	ReliefWeb, state EOC sources	Fills narrative and field-level details during unfolding events.
CDC + WHO	High-trust public health guidance and event framing for health crises.	Some updates are periodic rather than minute-by-minute.	Regional health agencies	Provides local confirmation and operational advisories.
GDACS	Global multi-hazard monitoring with practical alerting value.	Requires corroboration for casualty figures.	USGS/NOAA/FEMA or national authorities	Authoritative agencies verify event-specific factual details.
ReliefWeb	Strong humanitarian document stream and multi-agency reports.	Contains narrative reports that need verification against primary authorities.	Official government and UN agency bulletins	Ensures narrative claims map to authoritative fact signals.
NASA FIRMS	Strong remote-sensing support for wildfire detection at scale.	Signal source, not full incident command status.	NIFC/Cal Fire/state fire agencies	Operational agencies validate fire status, containment, and actions.

Source	Why Selected	Limitations	Backup / Reference Source	Why Backup is Useful
Websites considered but not primary: social media streams, crowdsourced dashboards, generic news blogs	Useful for early signal discovery only.	Higher misinformation/noise risk and inconsistent verification.	Use as `unverified lead indicators` only	Can speed detection but never treated as source-of-truth without corroboration.

Step 3 - Validation and Source-of-Truth Model

3.1 Source Tiers

- **Tier A (Authoritative):** Official government, intergovernmental, and designated emergency agencies.
- **Tier B (Institutional):** Reputable organizations and established operational partners.
- **Tier C (Signal-only):** Social/crowdsourced/unofficial reports; treated as unverified until corroborated.

3.2 Status Classification Rules

- **confirmed:** supported by Tier A, or by multiple independent high-credibility sources with no unresolved conflict.
- **preliminary:** credible early report with partial corroboration or evolving details.
- **unverified:** single-source signal, conflicting claims unresolved, or insufficient source quality.

3.3 Conflict Detection Rules

- Flag discrepancies in casualties, injuries, displaced counts, or affected area.
- Flag location/time mismatches for the same incident reference.
- Preserve all claims and link each to source + timestamp for auditability.

3.4 Source-of-Truth Principle

No single source is absolute for all fields. Source-of-truth is field-specific and evidence-weighted, with Tier A precedence and explicit conflict marking.

Step 4 - Why These Sources Are Strong for This Project

- **Authority and trust:** The selected stack prioritizes official agencies and internationally recognized institutions, minimizing misinformation risk.
- **Operational relevance:** Sources provide real event alerts, situational indicators, and response context aligned to catastrophic crisis workflows.
- **Coverage depth:** Combined sources cover seismic, weather, wildfire, humanitarian, and public-health events without over-relying on one domain.
- **Traceability:** Each selected source supports citation, timestamping, and provenance, enabling transparent downstream briefing generation.
- **Resilience:** The backup/reference strategy ensures continuity when one source is delayed, partial, or temporarily unavailable.
- **Verification-ready:** The source mix supports deterministic classification into confirmed, preliminary, and unverified states.

Conclusion: This source strategy is designed for high-integrity crisis intelligence ingestion where accuracy, timeliness, provenance, and conflict transparency are non-negotiable.